PRANJAL SINGH SESODIA

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EDUCATION

Netaji Subhas University of Technology (NSUT), ***formerly Netaji Subhas Institute of Technology*** **Aug 2024 - Aug 2028 (Expected)**
B.Tech. Mechanical Engineering | New Delhi, India *CGPA 8.07/10 (Dept. Rank 19/198, top 10%)*

- **Coursework:** Thermodynamics, Theory of Machines, Solid Mechanics, Fluid Mechanics, Materials Science, Manufacturing Processes, Metrology, Control Systems, Numerical Methods; ongoing: Machine Design, Robotics

EXPERIENCE

R&D Intern, mPRAGATI, Indian Institute of Technology Delhi (IIT Delhi) **Jun 2026 - Present**
New Delhi, India | Supervisor: Dr. Dinesh Kalyanasundaram

- Built a **6-DOF robotic arm** hardware prototype for end-effector stabilization despite visual occlusions
- Achieved **encoder-free closed-loop servo control** via eye-in-hand/eye-to-hand vision-based state estimation
- Optimized a patented elbow implant for manufacturability; ran **structural FEA** (bending), pull-out and bone-implant interface micromotion analyses; used multibody dynamics to show that the **actual ulnar trajectory** deviates from **the patented path**
- Redesigned a polycentric four-bar knee linkage to satisfy the Grashof condition and optimize clearances, **restoring full flexion and extension** in physical testing
- Designed a 5:1 reduction **cycloidal gearbox**, performed a motion study, and tested a 3D-printed prototype; prepared **manufacturing drawings with GD&T**

Research Intern, QuEST Lab, IIT Delhi **Dec 2025 - May 2026**
New Delhi, India | Supervisor: Dr. Jay Dhariwal

- Co-developed a pump and lining system for saline management in **minimally invasive surgery (MIS)**
- Built a webcam-based stereo vision rig (7 cm baseline) for markerless surgeon motion measurement, achieving **95-96% depth accuracy** against test-jig reference distances using stereo triangulation
- Built joint-angle tracking (AlphaPose) with limb-line extrapolation to **handle occlusions**; ran Depth Anything at ~11 FPS

PUBLICATION

- **Sesodia, P. S.**, & Dabass, V. (2026). *Robust Real-Time Levitation Gap Estimation for Hyperloop Pods*. Next Research (Elsevier). DOI: 10.1016/j.nexres.2026.102384

ENGINEERING PROJECTS

Tabletop Subsonic Wind Tunnel **Mar 2026 - Present**
NSUT Delhi | Supervisor: Dr. Vivek Kumar

- Designed a **150 mm test section and 6:1 contraction nozzle** using the Bell-Mehta curve; optimized internal flow geometry through CFD analysis in Ansys Fluent
- Reduced wall thickness and component mass through static structural **FEA**; integrated the drive and measurement electronics

Team Daedalus Racing, BAJA SAEINDIA **Nov 2025 - Mar 2026**
NSUT Delhi

- Optimized drivetrain gearing by calculating and implementing a reduction-ratio change from 11:1 to 8:1 for low-gradient endurance conditions, increasing theoretical output speed by **37.5%** at equal engine RPM
- Assessed low-cost catalysts and emissions for the catalytic converter (Ansys Granta Selector); co-presented the team's sustainability report and secured **1st place** nationally

EYEAR, Speech-to-Sign Device **2021 - 2022**

- Built a real-time speech-to-sign device (Raspberry Pi, Python); **presented to the Union Minister of Education**; Indian provisional patent filed (2022)

AWARDS & LEADERSHIP

- **1st place nationally (INR 30,000)**, mBAJA 2026 Sustainability Event, Team Daedalus Racing
- **Chhatra Pariyojana Vittiya Sahayata (CPVS) grant (INR 10,000)**, NSUT, for the tabletop wind tunnel (2026)
- Instructed **60 students** in a 2-day hands-on 3D-printing workshop, NSUT Innovation and Incubation Foundation (Sep 2026); mentored **8 students** to a Li-ion battery upcycling prototype in 3 weeks, Change.Makers 2.0, IIT Delhi (2025)

SKILLS

Mechanical Design & Fabrication: Geometric Dimensioning and Tolerancing (GD&T), Technical Drawings, Design for Manufacturing (DFM), Gearbox Design, Linkage Synthesis, Kinematic Analysis, Prototype Assembly and Testing, Rapid Prototyping (FDM/SLA 3D Printing)
CAD & Simulation: SolidWorks (parts, assemblies, drawings), Finite Element Analysis (Ansys Static Structural), Computational Fluid Dynamics (Ansys Fluent), Ansys Multibody Dynamics, Ansys Granta Selector, MuJoCo
Controls & Programming: ROS 2, PID Control, Computer Vision, Python (NumPy, OpenCV), MATLAB, C, Arduino, ESP32, Raspberry Pi
Languages: English (fluent), Hindi (native)